

Use Case Name: Create a science plan	ID: 1	Importance Level: High
Primary Actor: Astronomer		Use Case Type: Detail & Essential
Stakeholders and Interests: Science Observer		
Brief Description: This use case describes how science is created.		
Trigger: Astronomers need to propose a new science plan for collecting astronomical data.		
Type: External		
Relationships: Association: Astronomer Include: Extend: Generalization:		
Normal Flow of Events: 1. The system navigates the user to the science plan page. 2. The astronomer selects to create a science plan. 3. The system provides a template that the astronomer can fill in. 4. The system asks the astronomer to add their collaborator to this science plan from list of astronomer. 5. The system asks the astronomer to fill in the funding. 6. The system asks the astronomer to fill in the objective. 7. The system asks astronomers to choose the star system from the list. 8. The system asks the astronomer to select the date and time which is available in the schedule. 9. The system asks the astronomer to choose the telescope location. 10. The system asks the astronomer to select the data processing requirements from the list. 11. The system validates all the significant details for creating a new science plan. 12. The system provides generating status to astronomers.		
Subflows:		
Alternate/Exceptional Flow: 4, a1. The astronomer may not have the collaborators. - a2. The astronomer leaves this empty and jumps to the next step. 11, e1. The astronomer does not fill in all the significant details. - e2. The system displays an error message.		

Use Case Name: Submit Science Plan	ID:2	Importance Level: High
Primary Actor: Astronomer		Use Case Type: Detail & Essential
Stakeholders and Interests: Science observer		
Brief Description: This use case describes how a science plan is submitted.		
Trigger: Astronomers need to submit a new science plan for collected astronomical data		
Type: External		
Relationships: Association: Astronomer Include: Extend: Generalization:		
Normal Flow of Events: 1. The system navigates the user to the science plan page. 2. The system shows the astronomer's list of science plans including its status. 3. The astronomer selects to list down the tested science plan. 4. The system provides a list of tested science plans. 5. The astronomer selects to submit the science plan to the system. 6. The system provides the information of the science plan. 7. The astronomer selects to confirm to submit the science plan. 8. The system displays the status message back to the astronomers.		
Subflows:		
Alternate/Exceptional Flow: 7, a1. The astronomer alternatively wants to modify the science plan information. a2. The system cancels the test result and requires a rerun of the test.		

Use Case Name: Validate a science plan	ID: 3	Importance Level: High
Primary Actor: Science Observer		Use Case Type: Detail & Essential
Stakeholders and Interests: Science Observer - want to make sure the science plan work or not Astronomer - want to use their science plan to be the observing program		
Brief Description: This use case describes how science observer validate the science plan that created by an astronomer to further transform it to observing program.		
Trigger: When astronomer creates science plan Type: External		
Relationships: Association: Science Observer Include: Extend: Generalization:		
Normal Flow of Events: 1. The system validates the user's level of access. 2. The user goes to the submitted science plan page. 3. The user selects a science plan to be validated. 4. The user will do the review to ensure that it is scientifically sound and feasible, including funding, objective, star system, schedule, telescope location, and data constraint. 5. If the science plan passes all validation users will mark it as validated or else add feedback/comment and mark it as not pass. 6. The science plan was saved and updated in the system.		
Subflows:		
Alternate/Exceptional Flow: 1. Submitted science plan is empty 2. System failure occurs		